Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[utf8, twoside,]article

[utf8]inputenc standalone pgfplots compat=1.15 tikz import amsmath csquotes [american,spanish]babel biblatex-chicago csquotes graphicx setspace

mybib.bib

Title1

Título1 Nombre Apellido, *Institución Educativa*

Abstract

Abstract text in English

Resumen

Abstract text in Spanish

Keywords— one, two, three
Palabras clave— uno, dos, tres

1 Introduction

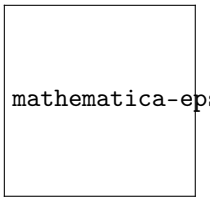
Introduction text. An example on how to cite:

Lorem ipsum dolor sit amet, consectetur adipiscing elit.[1] Pellentesque tincidunt sit amet magna a ultricies. Praesent ultrices dictum velit, ut luctus sem fermentum a.[2] Aenean sed ex fringilla nulla mattis cursus. Sed efficitur ligula condimentum purus vestibulum placerat.[3]

2 Body

Body text. The name of this section may be changed, and additional sections may be created.

.eps image:



?figurename? 1: eps image

An image made with pgfplots:

figs/graph.tex

?figurename? 2: pgfplots image

Example of a TikZ image:

figs/graph2.tex

?figurename? 3: TikZ image

Use the method most suited for your needs.
You may also reference images and tables. For example, Figure 1is a .eps image
A regression table
Maths example
Slustky's substitution matrix is:

$$\delta^2 e(p.u)\delta p_1^2 \delta^2 e(p.u)\delta p_2 \delta p_1 \delta^2 e(p.u)\delta p_1 \delta p_2 \delta^2 e(p.u)\delta p_2^2 \tag{1}$$

By the Shepard Theorem we know that

[1] [?]
[2] [?]
[3] [?]

tables/reg.tex

?tablename? 1: Table imported from Stata

$$\delta e(\vec{p}, u)\delta p_i = x_i^h(\vec{p}, u) \tag{2}$$

Donde $x_i^h(\vec{p}, u))$ es la demanda Hicksiana,
Ahora, por eq:2, podemos escribir eq:1 como:

$$\delta x_1^h(\vec{p}, u)\delta p_1 \delta x_1^h(\vec{p}, u)\delta p_2 \delta x_2^h(\vec{p}, u)\delta p_1 \delta x_2^h(\vec{p}, u)\delta p_2 \tag{3}$$

Hicks' Third Law states:
$\sum_{j=1}^{n} \delta x_i^h(\vec{p}, u)\delta p_j p_j = 0, \quad i = 1, \ldots, n$
Therefore,
$\delta x_1^h(\vec{p}, u)\delta p_1 \delta x_1^h(\vec{p}, u)\delta p_2$

$\delta x_2^h(\vec{p}, u)\delta p_1 \delta x_2^h(\vec{p}, u)\delta p_2$

p1

p_2

$= 0$

0

3 Conclusions

Conclusions text.